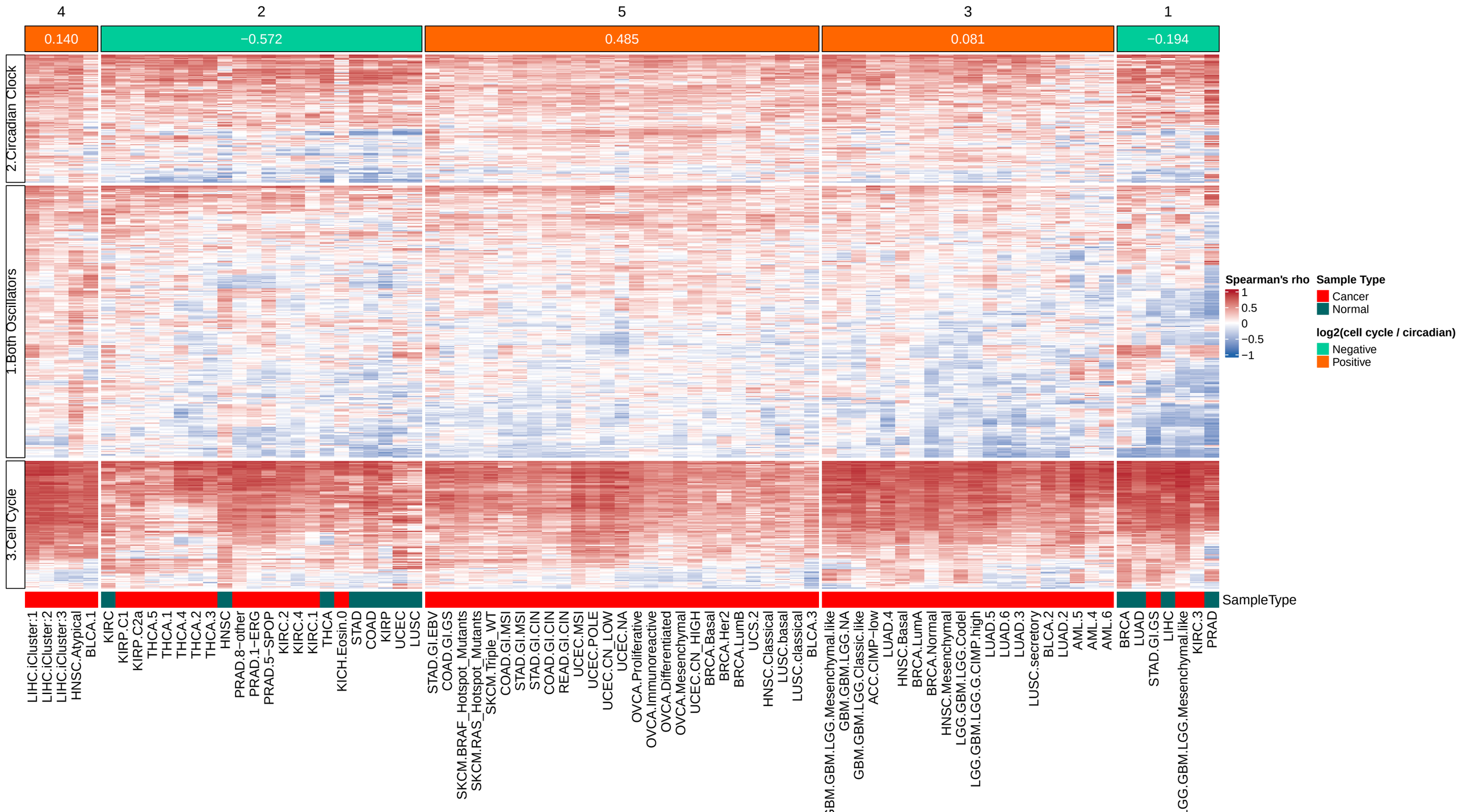
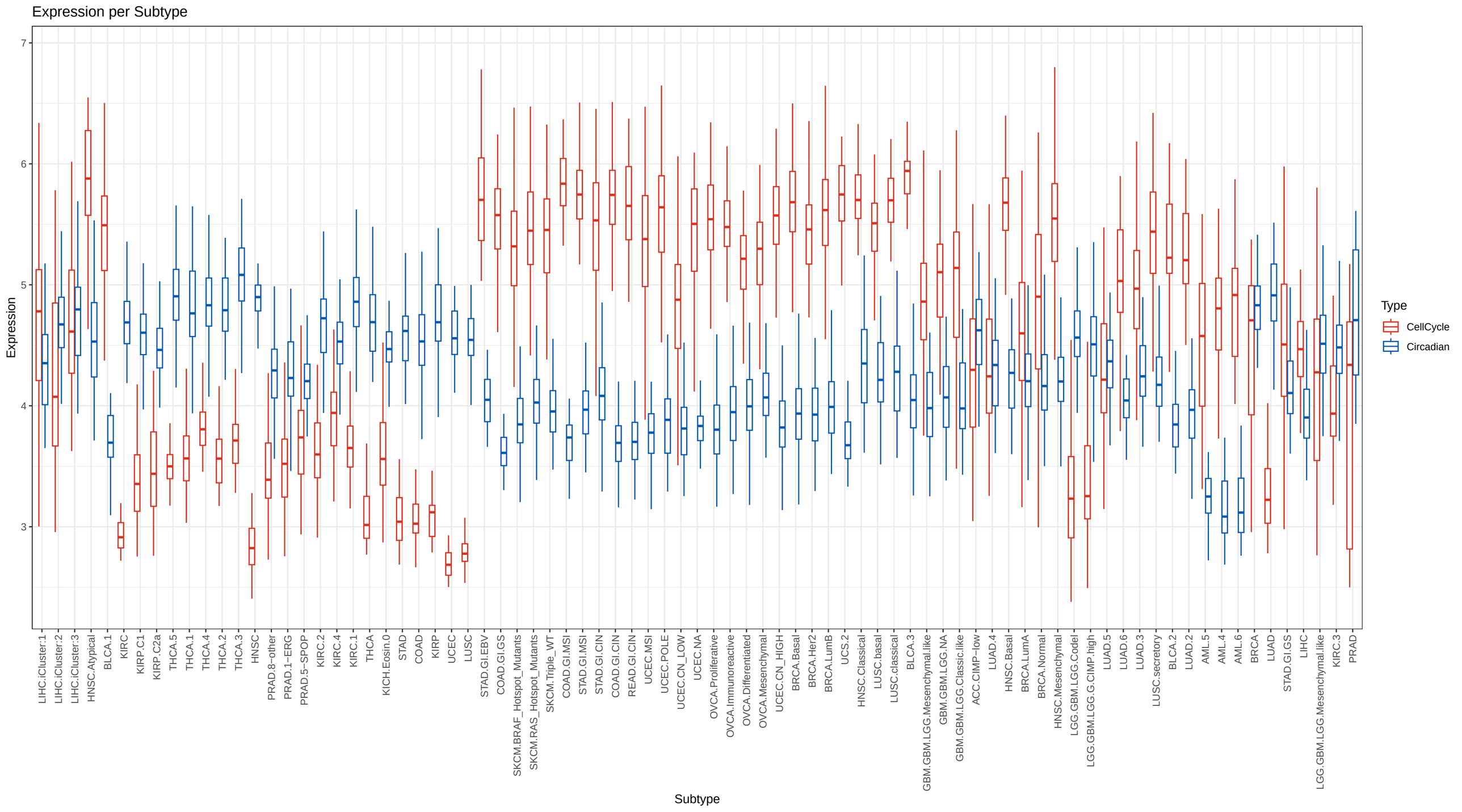
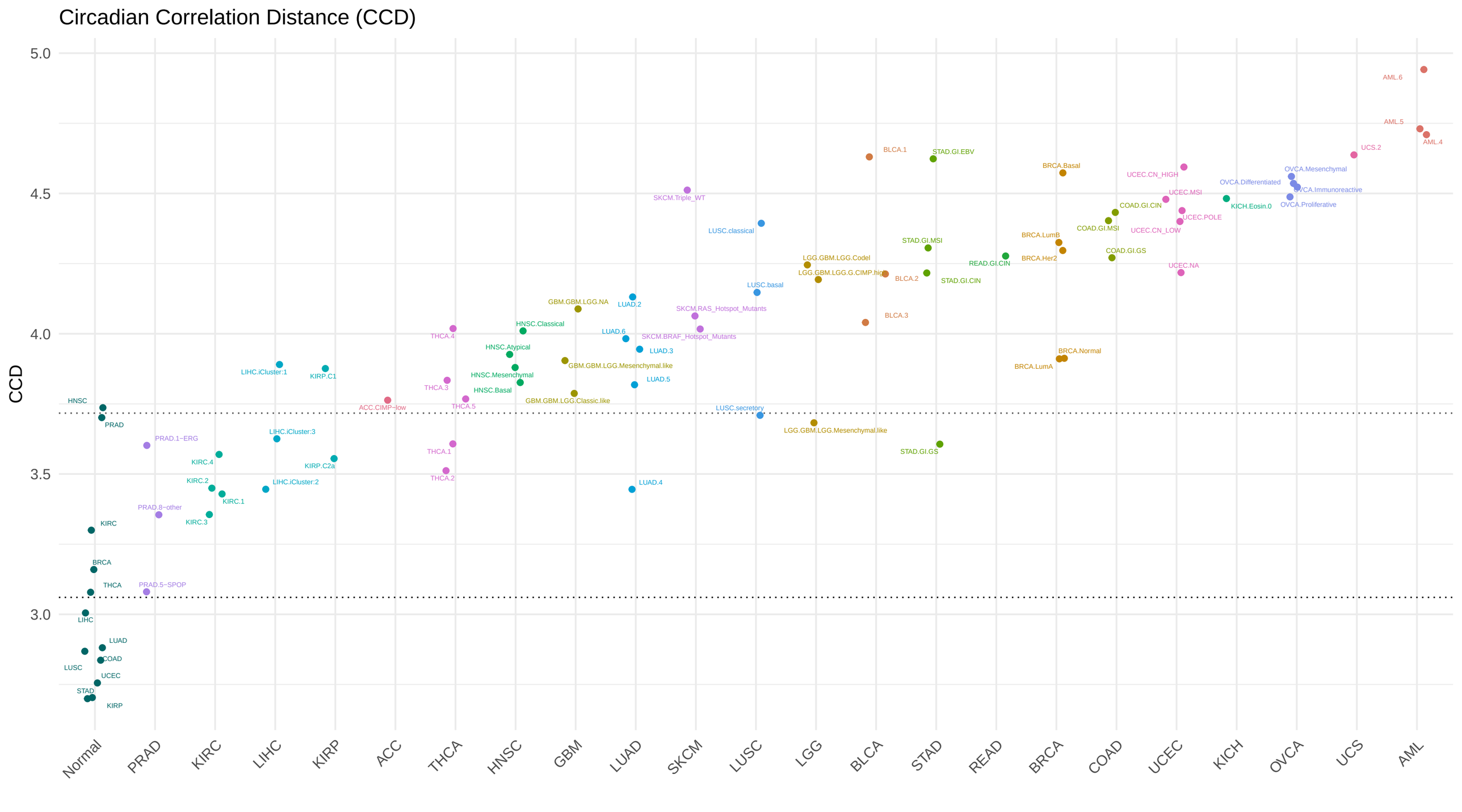
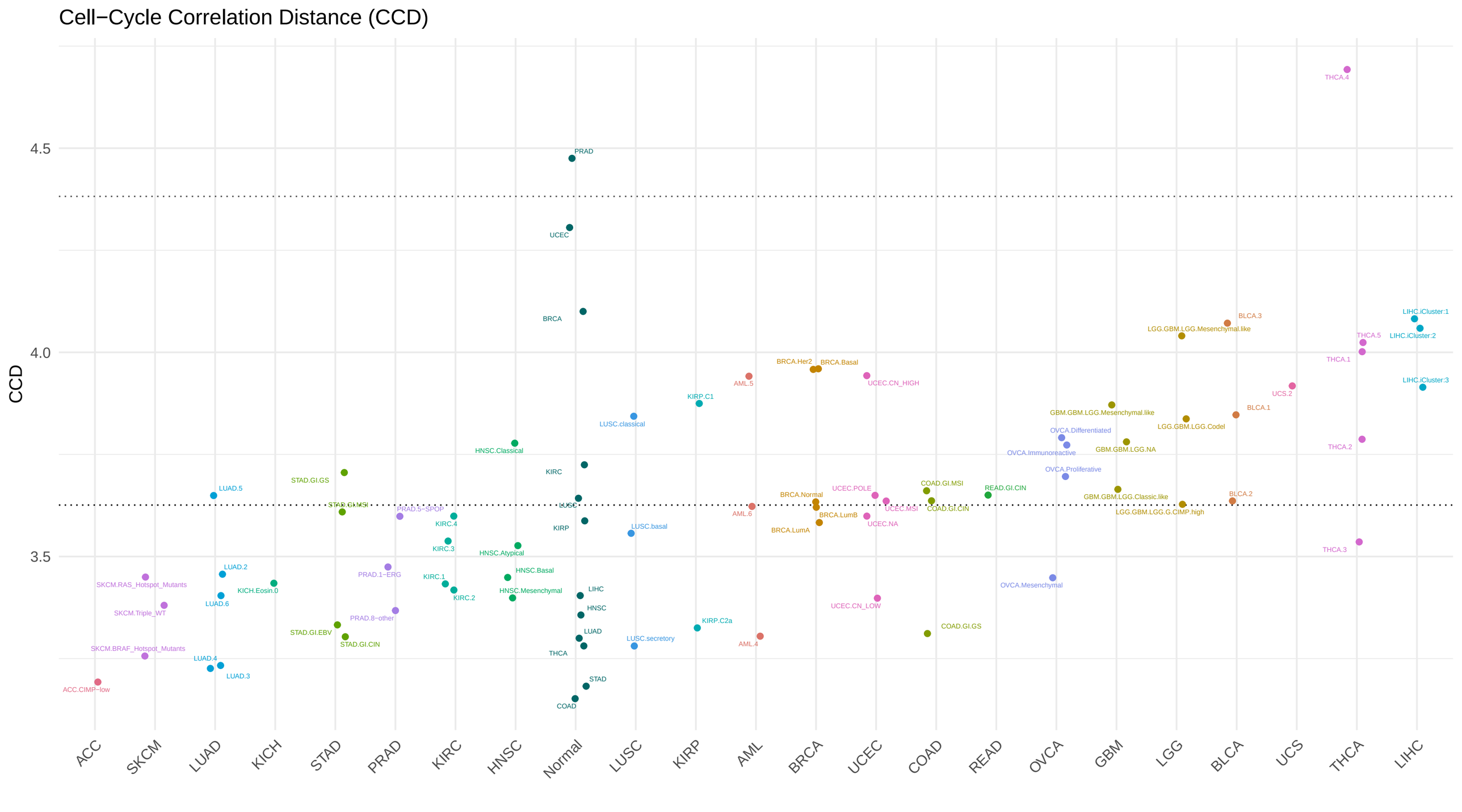
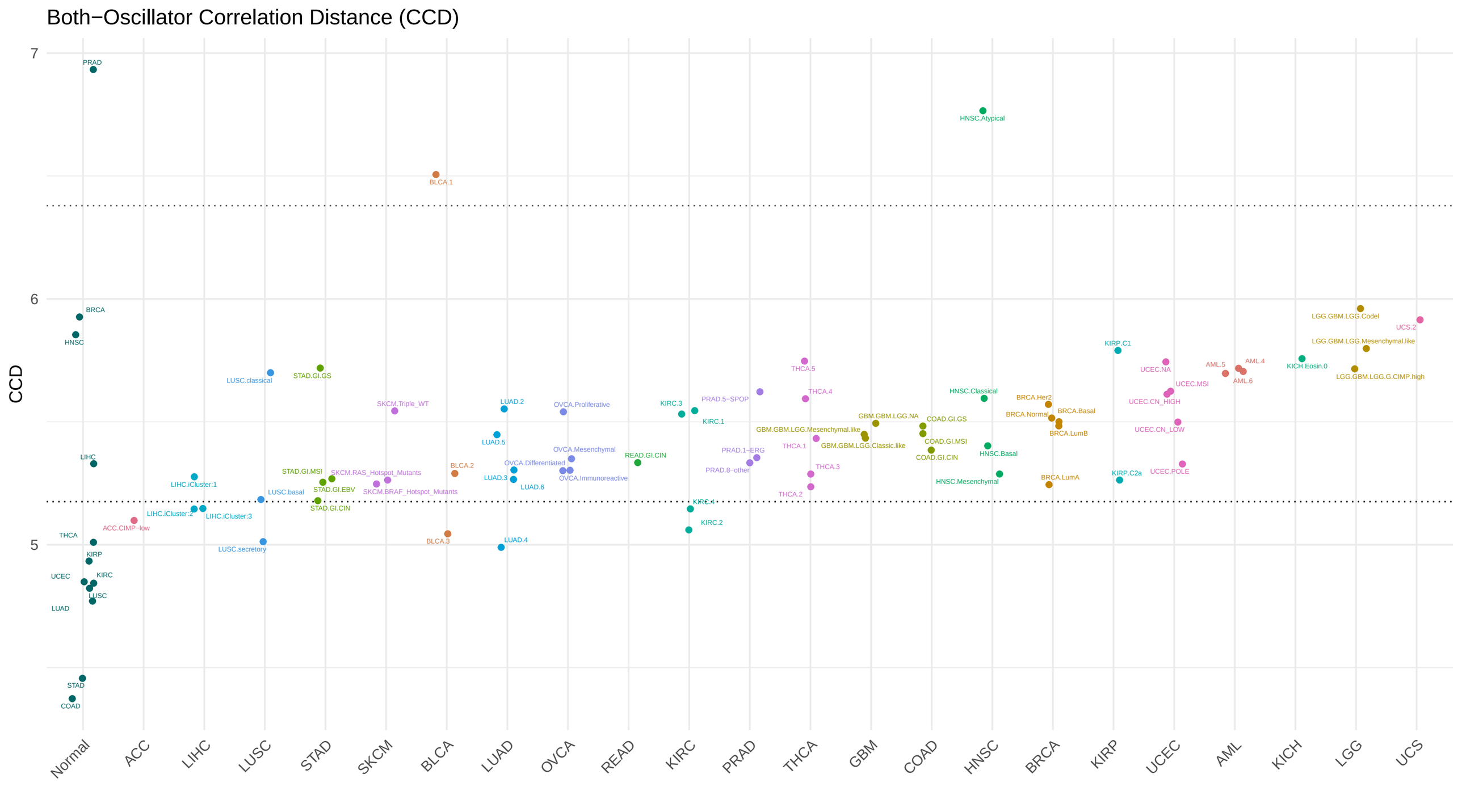
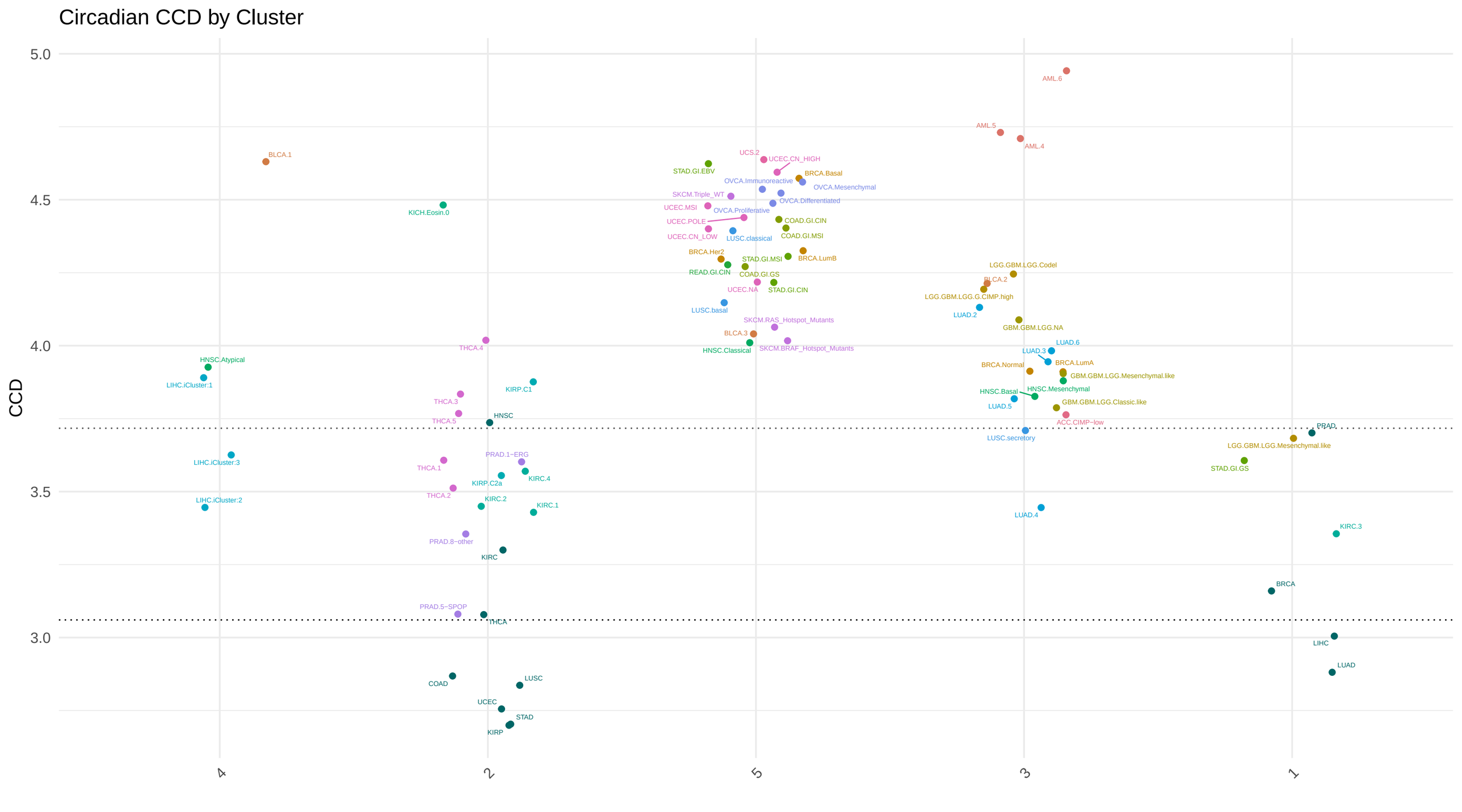

4
2
5
3
1
0.140
−0.572
0.485
0.081
−0.194
2.Circadian Clock
1.Both Oscillators
3.Cell Cycle
Spearman's rho
1
0.5
0
−0.5
−1
Sample Type
Cancer
Normal
log2(cell cycle / circadian)
Negative
Positive
SampleType
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
BRCA
LUAD
STAD.GI.GS
LIHC
LGG.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD

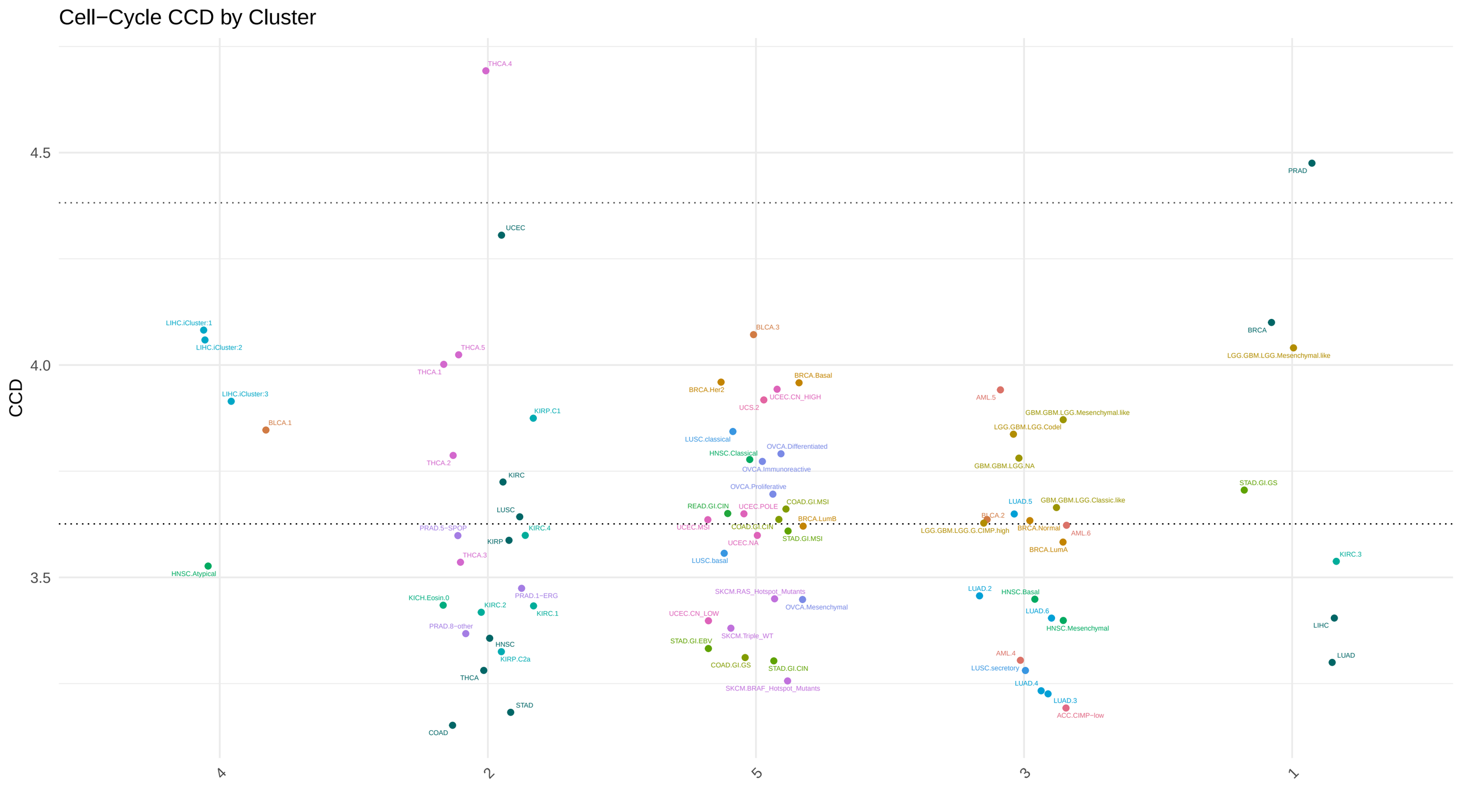

Expression per Subtype
Expression
7
6
5
4
3
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
BRCA
LUAD
STAD.GI.GS
LIHC
LGG.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD
Subtype
Type
CellCycle
Circadian

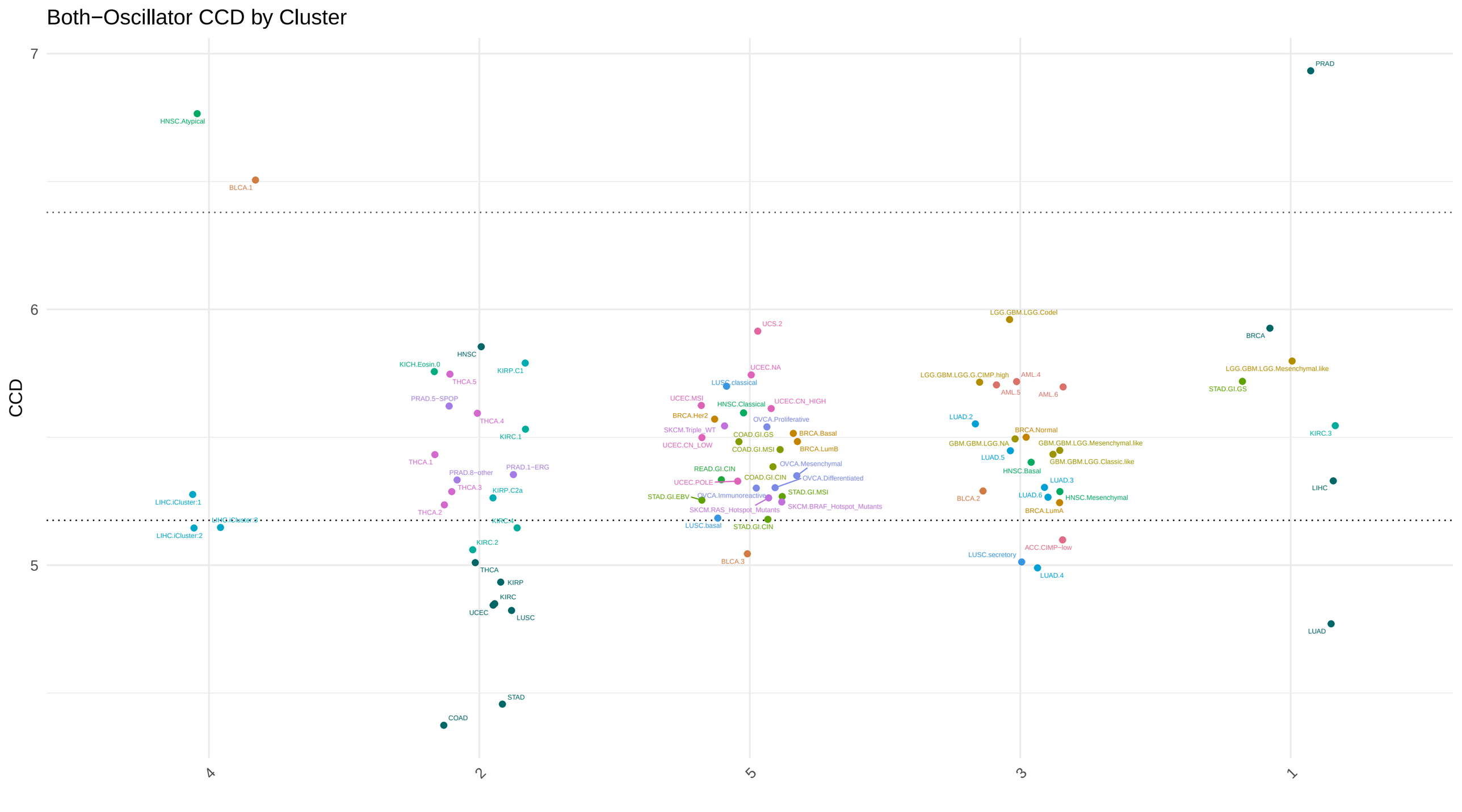

Circadian Correlation Distance (CCD)
CCD
5.0
4.5
4.0
3.5
3.0
Normal
PRAD
KIRC
LIHC
KIRP
ACC
THCA
HNSC
GBM
LUAD
SKCM
LUSC
LGG
BLCA
STAD
READ
BRCA
COAD
UCEC
KICH
OVCA
UCS
AML
HNSC
PRAD
PRAD.1−ERG
PRAD.8−other
PRAD.5−SPOP
KIRC
KIRC.4
KIRC.2
KIRC.1
KIRC.3
BRCA
THCA
LIHC
LUAD
COAD
LUSC
UCEC
STAD
KIRP
LIHC.iCluster:1
KIRP.C1
LIHC.iCluster:3
KIRP.C2a
LIHC.iCluster:2
ACC.CIMP−low
THCA.4
THCA.3
THCA.5
THCA.1
THCA.2
HNSC.Classical
HNSC.Atypical
HNSC.Mesenchymal
HNSC.Basal
GBM.GBM.LGG.NA
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.Classic.like
LUAD.2
LUAD.6
LUAD.3
LUAD.5
LUAD.4
SKCM.Triple_WT
SKCM.RAS_Hotspot_Mutants
SKCM.BRAF_Hotspot_Mutants
LUSC.classical
LUSC.basal
LUSC.secretory
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LGG.GBM.LGG.Mesenchymal.like
BLCA.1
BLCA.2
BLCA.3
STAD.GI.EBV
STAD.GI.MSI
STAD.GI.CIN
STAD.GI.GS
READ.GI.CIN
BRCA.Basal
BRCA.LumB
BRCA.Her2
BRCA.Normal
BRCA.LumA
COAD.GI.CIN
COAD.GI.MSI
COAD.GI.GS
UCEC.CN_HIGH
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
KICH.Eosin.0
OVCA.Mesenchymal
OVCA.Differentiated
OVCA.Immunoreactive
OVCA.Proliferative
UCS.2
AML.6
AML.5
AML.4

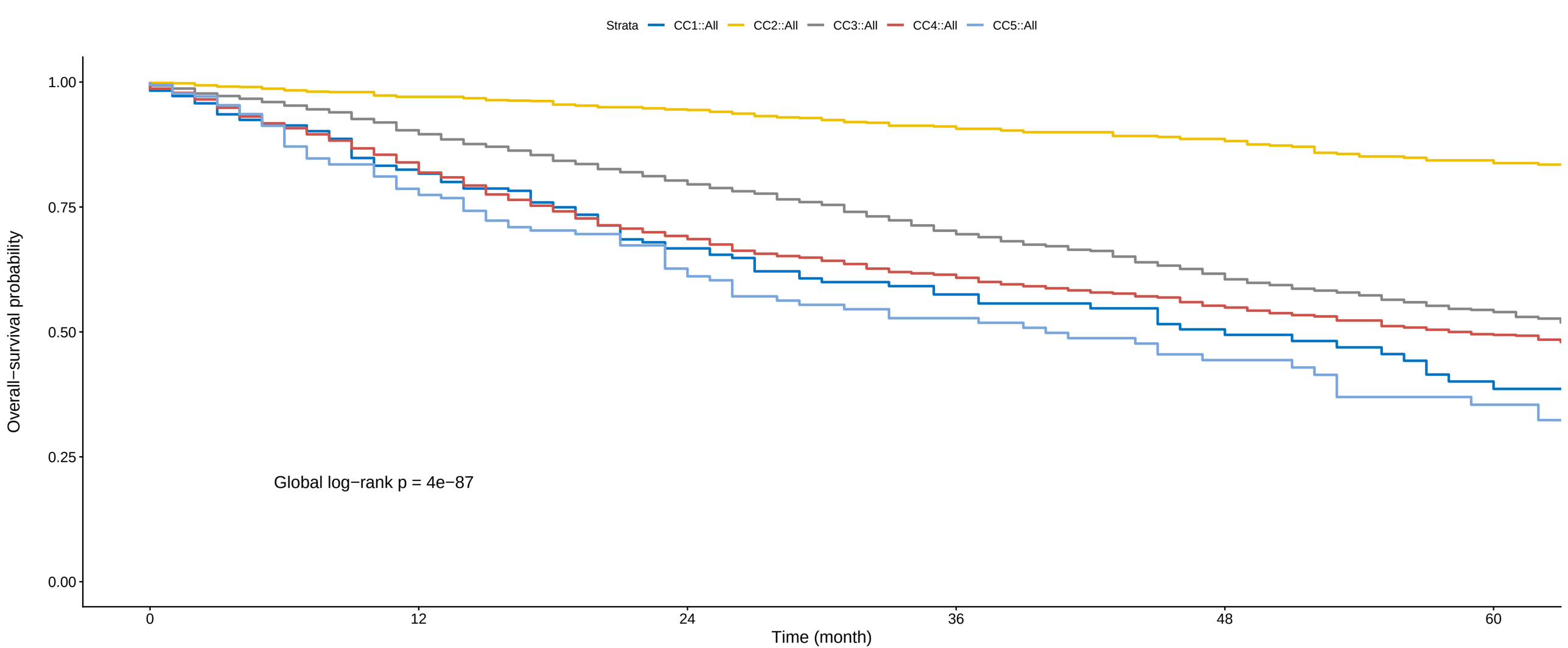

Cell−Cycle Correlation Distance (CCD)
CCD
4.5
4.0
3.5
ACC
SKCM
LUAD
KICH
STAD
PRAD
KIRC
HNSC
Normal
LUSC
KIRP
AML
BRCA
UCEC
COAD
READ
OVCA
GBM
LGG
BLCA
UCS
THCA
LIHC
ACC.CIMP−low
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
LUAD.4
LUAD.3
LUAD.5
LUAD.2
LUAD.6
KICH.Eosin.0
STAD.GI.EBV
STAD.GI.CIN
STAD.GI.GS
STAD.GI.MSI
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.4
KIRC.3
KIRC.1
KIRC.2
HNSC.Atypical
HNSC.Basal
HNSC.Mesenchymal
HNSC.Classical
COAD
STAD
THCA
LUAD
HNSC
LIHC
KIRP
KIRC
LUSC
PRAD
UCEC
BRCA
LUSC.secretory
LUSC.basal
LUSC.classical
KIRP.C2a
KIRP.C1
AML.4
AML.6
AML.5
BRCA.LumA
BRCA.LumB
BRCA.Normal
BRCA.Basal
BRCA.Her2
UCEC.CN_LOW
UCEC.NA
UCEC.MSI
UCEC.POLE
UCEC.CN_HIGH
COAD.GI.GS
COAD.GI.CIN
COAD.GI.MSI
OVCA.Mesenchymal
OVCA.Immunoreactive
OVCA.Proliferative
OVCA.Differentiated
READ.GI.CIN
GBM.GBM.LGG.Classic.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.G.CIMP.high
LGG.GBM.LGG.Codel
LGG.GBM.LGG.Mesenchymal.like
BLCA.2
BLCA.1
BLCA.3
UCS.2
THCA.3
THCA.2
THCA.1
THCA.5
THCA.4
LIHC.iCluster:3
LIHC.iCluster:2
LIHC.iCluster:1


Both−Oscillator Correlation Distance (CCD)
CCD
7
6
5
PRAD
BRCA
HNSC
LIHC
THCA
KIRP
UCEC
KIRC
LUSC
LUAD
STAD
COAD
HNSC.Atypical
BLCA.1
LUSC.classical
STAD.GI.GS
SKCM.Triple_WT
LUAD.2
OVCA.Proliferative
LUAD.5
KIRC.3
PRAD.5−SPOP
KIRC.1
THCA.5
THCA.4
HNSC.Classical
BRCA.Her2
BRCA.Basal
BRCA.Normal
BRCA.LumB
GBM.GBM.LGG.NA
COAD.GI.GS
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.Classic.like
COAD.GI.MSI
COAD.GI.CIN
THCA.1
HNSC.Basal
KIRP.C1
UCEC.NA
UCEC.MSI
UCEC.CN_HIGH
UCEC.CN_LOW
UCEC.POLE
KIRP.C2a
HNSC.Mesenchymal
BRCA.LumA
AML.5
AML.4
AML.6
KICH.Eosin.0
LGG.GBM.LGG.Codel
UCS.2
LGG.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.G.CIMP.high
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
ACC.CIMP−low
LUSC.basal
LUSC.secretory
STAD.GI.MSI
STAD.GI.EBV
STAD.GI.CIN
SKCM.RAS_Hotspot_Mutants
SKCM.BRAF_Hotspot_Mutants
BLCA.2
BLCA.3
OVCA.Differentiated
OVCA.Mesenchymal
OVCA.Immunoreactive
LUAD.3
LUAD.6
LUAD.4
READ.GI.CIN
KIRC.4
KIRC.2
PRAD.1−ERG
PRAD.8−other
THCA.3
THCA.2
Normal
ACC
LIHC
LUSC
STAD
SKCM
BLCA
LUAD
OVCA
READ
KIRC
PRAD
THCA
GBM
COAD
HNSC
BRCA
KIRP
UCEC
AML
KICH
LGG
UCS


Circadian CCD by Cluster
CCD
5.0
4.5
4.0
3.5
3.0
4
2
5
3
1
AML.6
AML.5
AML.4
BLCA.1
KICH.Eosin.0
UCS.2
UCEC.CN_HIGH
STAD.GI.EBV
BRCA.Basal
OVCA.Immunoreactive
OVCA.Mesenchymal
SKCM.Triple_WT
OVCA.Differentiated
UCEC.MSI
OVCA.Proliferative
UCEC.POLE
COAD.GI.CIN
UCEC.CN_LOW
LUSC.classical
COAD.GI.MSI
BRCA.Her2
STAD.GI.MSI
BRCA.LumB
READ.GI.CIN
COAD.GI.GS
UCEC.NA
STAD.GI.CIN
LUSC.basal
SKCM.RAS_Hotspot_Mutants
BLCA.3
HNSC.Classical
SKCM.BRAF_Hotspot_Mutants
LGG.GBM.LGG.Codel
BLCA.2
LGG.GBM.LGG.G.CIMP.high
LUAD.2
GBM.GBM.LGG.NA
LUAD.6
LUAD.3
BRCA.Normal
BRCA.LumA
GBM.GBM.LGG.Mesenchymal.like
HNSC.Basal
HNSC.Mesenchymal
LUAD.5
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUSC.secretory
PRAD
LGG.GBM.LGG.Mesenchymal.like
STAD.GI.GS
THCA.4
HNSC.Atypical
LIHC.iCluster:1
KIRP.C1
THCA.3
THCA.5
HNSC
LIHC.iCluster:3
PRAD.1−ERG
THCA.1
KIRC.4
KIRP.C2a
THCA.2
KIRC.2
KIRC.1
LIHC.iCluster:2
LUAD.4
KIRC.3
PRAD.8−other
KIRC
BRCA
PRAD.5−SPOP
THCA
LIHC
LUAD
COAD
LUSC
UCEC
STAD
KIRP

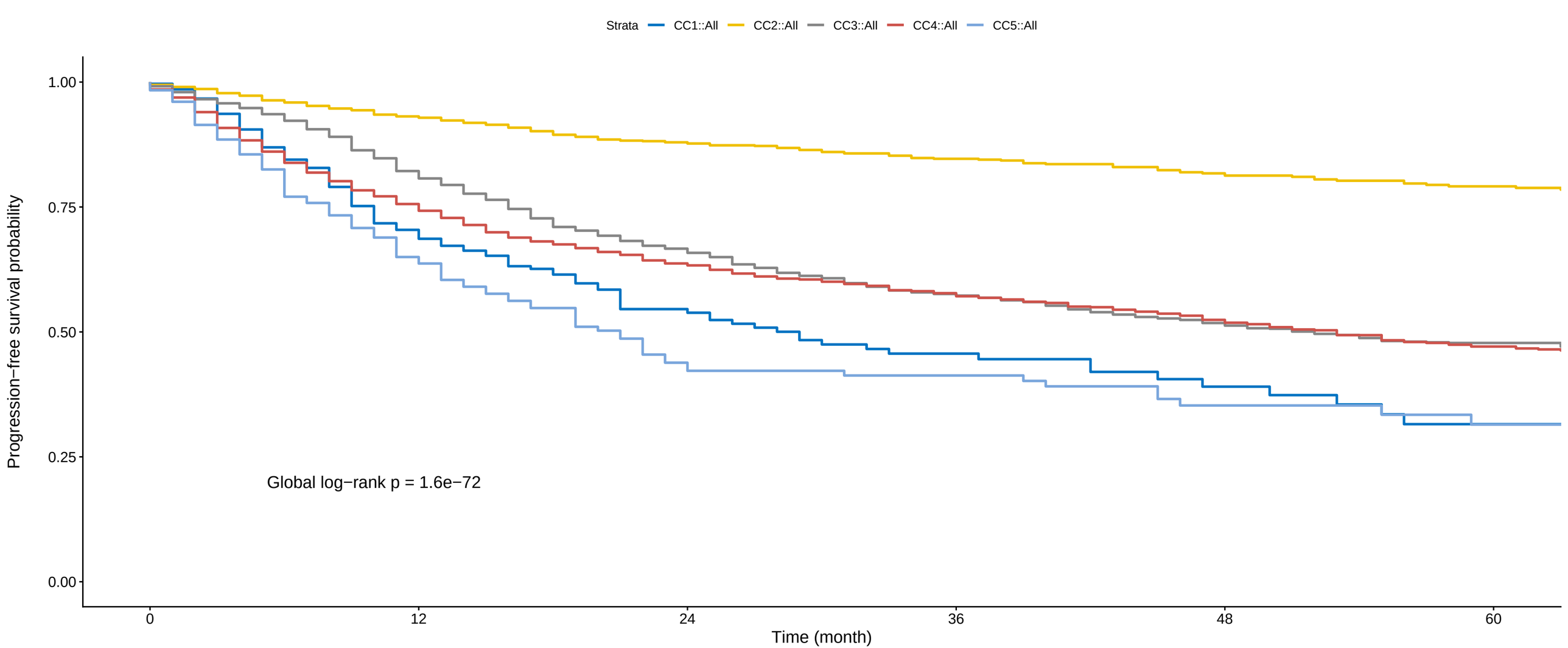

Cell−Cycle CCD by Cluster
CCD
4.5
4.0
3.5
4
2
5
3
1
THCA.4
PRAD
UCEC
LIHC.iCluster:1
LIHC.iCluster:2
THCA.5
THCA.1
LIHC.iCluster:3
BLCA.1
THCA.2
KIRP.C1
KIRC
LUSC
PRAD.5−SPOP
KIRC.4
KIRP
THCA.3
HNSC.Atypical
KICH.Eosin.0
PRAD.1−ERG
KIRC.2
KIRC.1
PRAD.8−other
HNSC
KIRP.C2a
THCA
STAD
COAD
BLCA.3
BRCA.Basal
BRCA.Her2
UCEC.CN_HIGH
UCS.2
LUSC.classical
OVCA.Differentiated
HNSC.Classical
OVCA.Immunoreactive
OVCA.Proliferative
READ.GI.CIN
UCEC.POLE
COAD.GI.MSI
BRCA.LumB
UCEC.MSI
COAD.GI.CIN
STAD.GI.MSI
UCEC.NA
LUSC.basal
SKCM.RAS_Hotspot_Mutants
OVCA.Mesenchymal
UCEC.CN_LOW
SKCM.Triple_WT
STAD.GI.EBV
COAD.GI.GS
STAD.GI.CIN
SKCM.BRAF_Hotspot_Mutants
BRCA
LGG.GBM.LGG.Mesenchymal.like
AML.5
GBM.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.Codel
GBM.GBM.LGG.NA
STAD.GI.GS
LUAD.5
GBM.GBM.LGG.Classic.like
BLCA.2
LGG.GBM.LGG.G.CIMP.high
BRCA.Normal
AML.6
BRCA.LumA
KIRC.3
LUAD.2
HNSC.Basal
LUAD.6
HNSC.Mesenchymal
LIHC
LUAD
AML.4
LUSC.secretory
LUAD.4
LUAD.3
ACC.CIMP−low


Both−Oscillator CCD by Cluster
CCD
7
6
5
4
2
5
3
1
PRAD
HNSC.Atypical
BLCA.1
LGG.GBM.LGG.Codel
UCS.2
BRCA
HNSC
KICH.Eosin.0
KIRP.C1
THCA.5
UCEC.NA
LUSC.classical
LGG.GBM.LGG.G.CIMP.high
AML.4
AML.5
AML.6
LGG.GBM.LGG.Mesenchymal.like
STAD.GI.GS
PRAD.5−SPOP
UCEC.MSI
HNSC.Classical
UCEC.CN_HIGH
BRCA.Her2
OVCA.Proliferative
LUAD.2
THCA.4
SKCM.Triple_WT
COAD.GI.GS
BRCA.Basal
BRCA.Normal
KIRC.1
UCEC.CN_LOW
COAD.GI.MSI
BRCA.LumB
GBM.GBM.LGG.NA
GBM.GBM.LGG.Mesenchymal.like
KIRC.3
THCA.1
LUAD.5
GBM.GBM.LGG.Classic.like
PRAD.8−other
PRAD.1−ERG
READ.GI.CIN
OVCA.Mesenchymal
HNSC.Basal
UCEC.POLE
COAD.GI.CIN
OVCA.Differentiated
LUAD.3
THCA.3
KIRP.C2a
LIHC
LIHC.iCluster:1
STAD.GI.EBV
OVCA.Immunoreactive
STAD.GI.MSI
BLCA.2
LUAD.6
HNSC.Mesenchymal
THCA.2
SKCM.RAS_Hotspot_Mutants
SKCM.BRAF_Hotspot_Mutants
BRCA.LumA
LIHC.iCluster:3
KIRC.4
LUSC.basal
STAD.GI.CIN
LIHC.iCluster:2
KIRC.2
ACC.CIMP−low
BLCA.3
LUSC.secretory
THCA
LUAD.4
KIRP
KIRC
UCEC
LUSC
LUAD
STAD
COAD



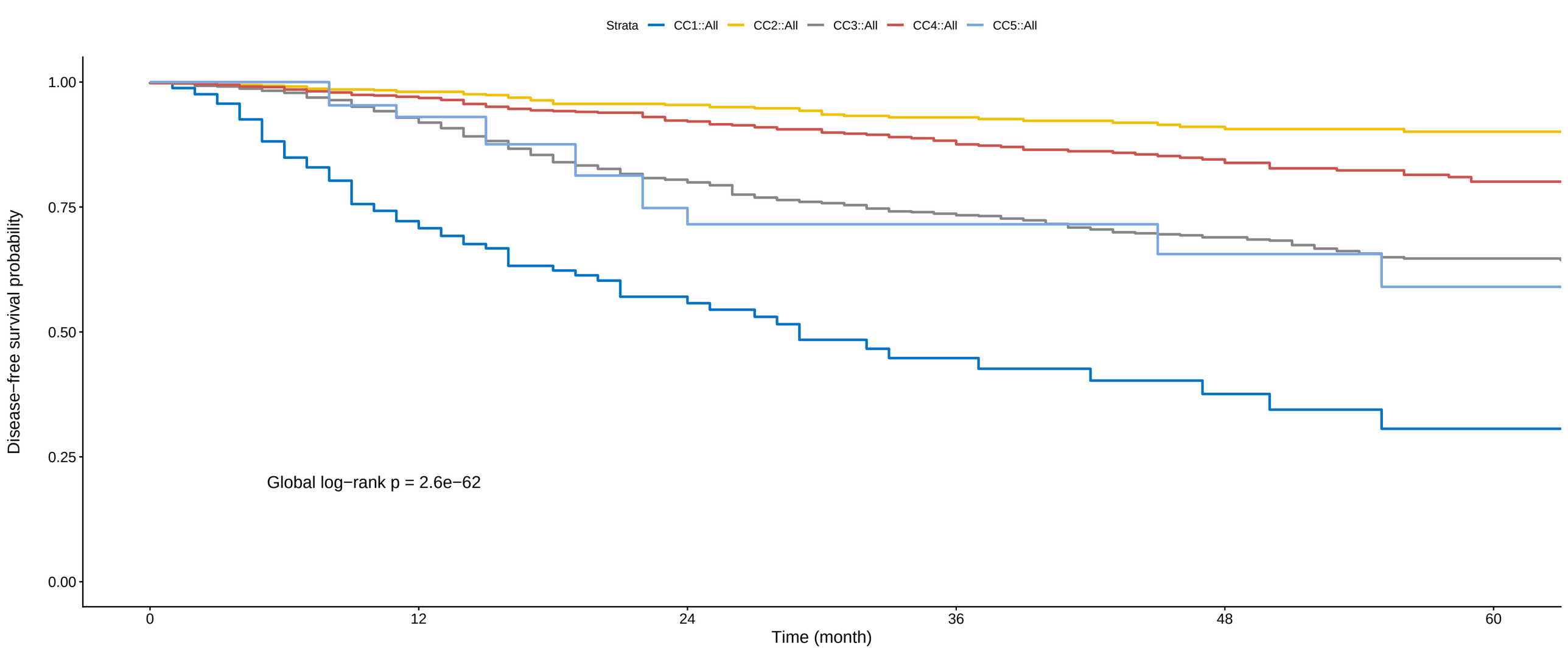
Number at risk

CC1::All	291	206	106	67	46	27
CC2::All	1243	1096	819	579	418	304
CC3::All	2746	2163	1448	984	711	513
CC4::All	2201	1609	986	683	454	331
CC5::All	182	128	81	57	36	23



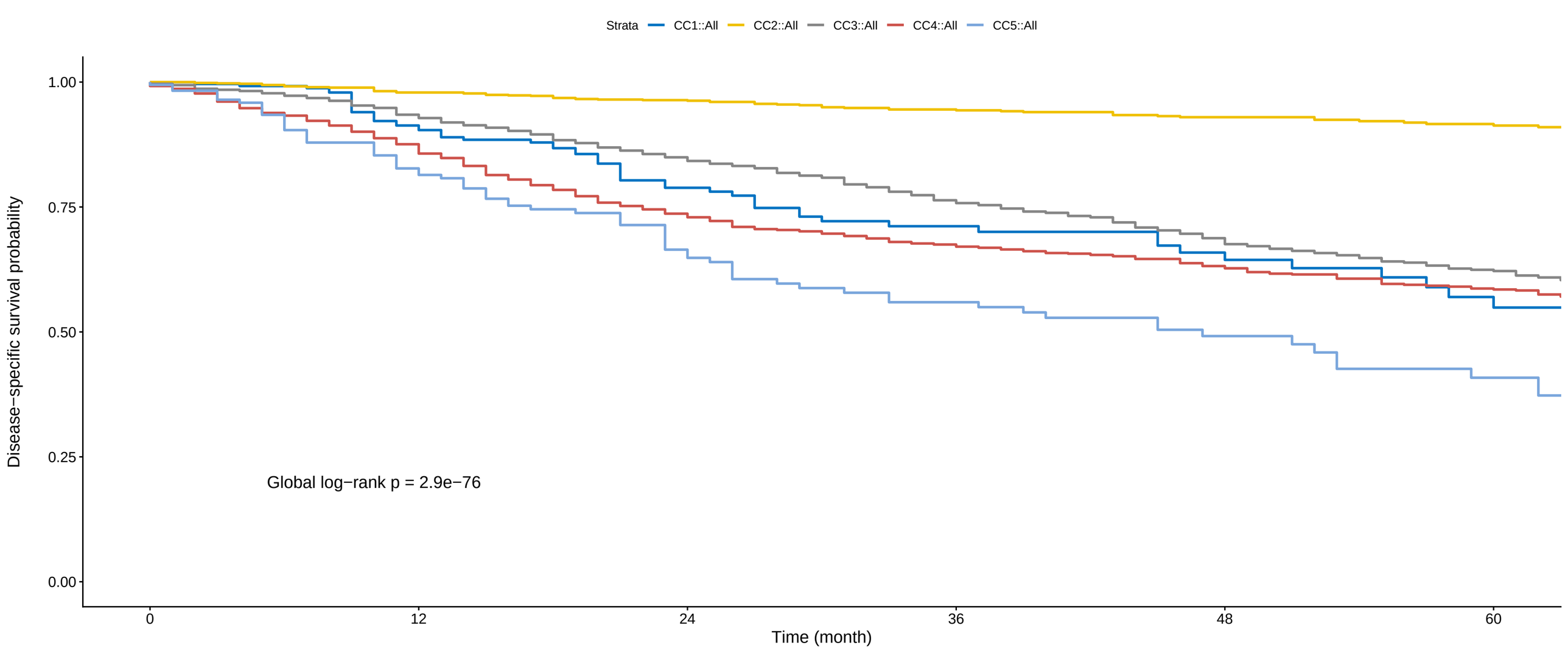
Number at risk

CC1::All	291	158	75	43	24	12
CC2::All	1240	1040	743	518	366	261
CC3::All	2748	1892	1117	701	500	353
CC4::All	2100	1345	814	563	367	254
CC5::All	183	100	54	42	24	16



Number at risk

CC1::All	177	102	44	22	13	6
CC2::All	690	629	449	291	204	153
CC3::All	1503	1211	731	465	329	231
CC4::All	915	767	501	357	244	172
CC5::All	46	40	23	17	11	8



Number at risk

CC1::All	280	199	103	66	45	27
CC2::All	1228	1083	809	573	412	299
CC3::All	2637	2091	1398	947	690	501
CC4::All	2011	1501	915	633	424	308
CC5::All	180	127	81	57	36	23